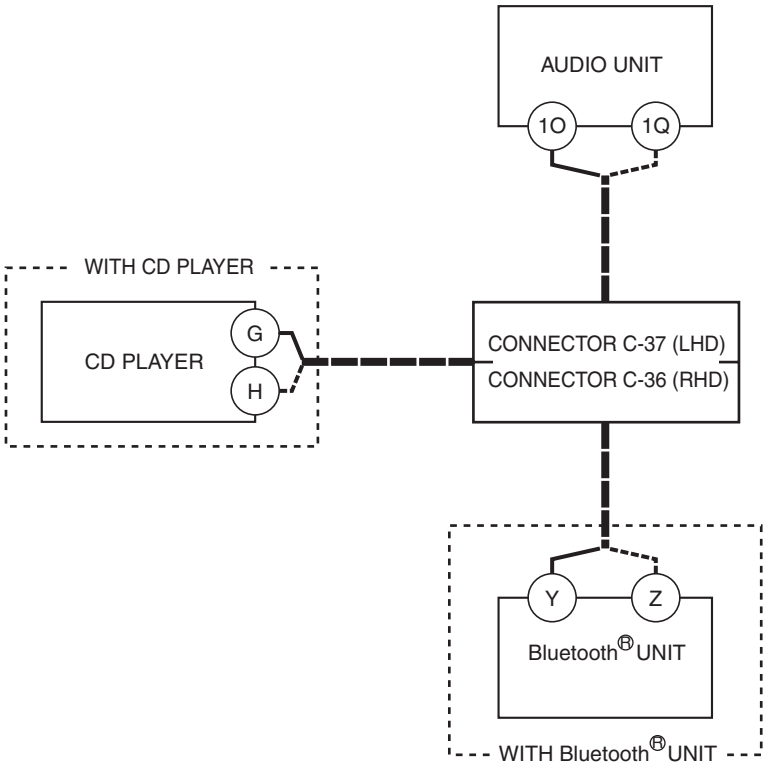
DTC: 16:Er11, 16:Er12, 26:Er81 [AUDIO]

id0902f6453700

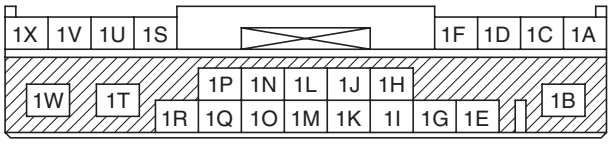
Description	• 16:Er11 — CAN communication time-out • 16:Er12 — Unit communication error • 26:Er81 — Communication error with Bluetooth® unit
Detection condition	• 16:Er11 — Time-out occurs in the audio unit CAN signal. • 16:Er12 — The audio unit cannot receive CAN signal. • 26:Er81 — The audio unit cannot receive CAN signal from the Bluetooth® unit.
Fail-safe	• 16:Er11 — Not applicable • 16:Er12 — Not applicable • 26:Er81 — Not applicable
Possible cause	• Audio unit connector or terminal malfunction • CD player connector or terminal malfunction (With CD player) • Bluetooth® unit connector or terminal malfunction (With Bluetooth® unit) • Connector C-37 connector or terminal malfunction (L.H.D.) • Connector C-36 connector or terminal malfunction (R.H.D.) • Short to power supply in wiring harness between the following terminals: — Audio unit terminal 1O and CD player terminal G (With CD player) — Audio unit terminal 1O and Bluetooth® unit terminal Y (With Bluetooth® unit) • Short to ground in wiring harness between the following terminals: — Audio unit terminal 1Q and CD player terminal H (With CD player) — Audio unit terminal 1Q and Bluetooth® unit terminal Z (With Bluetooth® unit) • Short between circuits in the following wiring harness: — Audio unit terminal 1O and CD player terminal G (With CD player) — Audio unit terminal 1Q and CD player terminal H (With CD player) — Audio unit terminal 1O and Bluetooth® unit terminal Y (With Bluetooth® unit) — Audio unit terminal 1Q and Bluetooth® unit terminal Z (With Bluetooth® unit) • Open circuit in wiring harness between the following terminals: — L.H.D.: • CD player terminal G and connector C-37 (With CD player) • CD player terminal H and connector C-37 (With CD player) • Bluetooth® unit terminal Y and connector C-37 (With Bluetooth® unit) • Bluetooth® unit terminal Z and connector C-37 (With Bluetooth® unit) • Audio unit terminal 1O and connector C-37 • Audio unit terminal 1Q and connector C-37 — R.H.D.: • CD player terminal G and connector C-36 (With CD player) • CD player terminal H and connector C-36 (With CD player) • Bluetooth® unit terminal Y and connector C-36 (With Bluetooth® unit) • Bluetooth® unit terminal Z and connector C-36 (With Bluetooth® unit) • Audio unit terminal 1O and connector C-36 • Audio unit terminal 1Q and connector C-36 • CD player malfunction (With CD player) • Bluetooth® unit malfunction (With Bluetooth® unit) • Audio unit malfunction

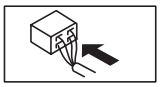
Description	• 16:Er11 — CAN communication time-out • 16:Er12 — Unit communication error • 26:Er81 — Communication error with Bluetooth® unit

▬▬▬ : INSTRUMENT PANEL HARNESS

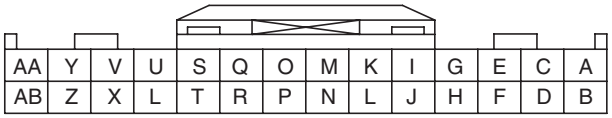


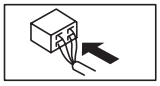
AUDIO UNIT
WIRING HARNESS-SIDE CONNECTOR



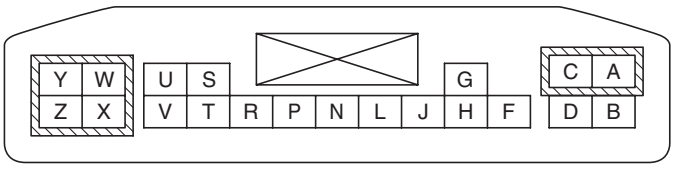


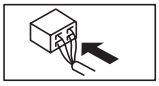
Bluetooth® UNIT
WIRING HARNESS-SIDE CONNECTOR





CD PLAYER
WIRING HARNESS-SIDE CONNECTOR





Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT FOR POWER SUPPLY-RELATED MALFUNCTION** • Switch the ignition ON (engine off). • Perform the DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [i-ELOOP].) • Is the power supply or DC-DC converter-related DTC displayed?	Yes	**Note** • It may be affected by power supply-related or DC-DC converter malfunction. Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
2	**INSPECT CAN_H SIDE CIRCUIT OF LOCAL-CAN FOR SHORT TO POWER SUPPLY** • Measure the voltage between audio unit terminal 1O and body ground. • Is the voltage **B+**?	Yes	A short to power supply in CAN_H side of LOCAL-CAN has occurred. • Go to Step 5.
		No	Go to the next step.
3	**INSPECT CAN_L SIDE CIRCUIT OF LOCAL-CAN FOR SHORT TO GROUND** • Measure the voltage between audio unit terminal 1Q (LOCAL-CAN CAN_L side) and body ground. • Is the voltage **0 V**?	Yes	A short to ground in CAN_L side of LOCAL-CAN has occurred. • Go to Step 6.
		No	Go to the next step.
4	**INSPECT BETWEEN LOCAL-CAN_L AND LOCAL-CAN_H SIDE CIRCUITS FOR SHORT BETWEEN CIRCUITS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the resistance between audio unit 1O and 1Q terminals (wiring harness-side). • Is the resistance **approx. 60 ohms**?	Yes	Go to Step 9. (LOCAL-CAN is normal)
		No	**If the resistance is 0 ohm:** • A short between circuits has occurred in LOCAL-CAN. — Go to Step 7. **If the resistance is 120 ohms:** • An open circuit in the LOCAL-CAN has occurred. — Go to Step 8.
5	**INSPECT RELATED WIRING HARNESS (SHORT TO POWER SUPPLY)** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the following connectors: — Audio unit — CD player (With CD player) — Bluetooth® unit (With Bluetooth® unit) • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side). — Audio unit terminal 1O — CD player terminal G (With CD player) — Bluetooth® unit terminal Y (With Bluetooth® unit) • Is the voltage **0 V**?	Yes	Go to Step 9.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Audio unit terminal 1O and CD player terminal G (With CD player) • Audio unit terminal 1O and Bluetooth® unit terminal Y (With Bluetooth® unit) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 11.

Step	Inspection		Action
6	**INSPECT RELATED WIRING HARNESS (SHORT TO GROUND)** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the following connectors: — Audio unit — CD player (With CD player) — Bluetooth® unit (With Bluetooth® unit) • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Audio unit terminal 1Q — CD player terminal H (With CD player) — Bluetooth® unit terminal Z (With Bluetooth® unit) • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Audio unit terminal 1Q and CD player terminal H (With CD player) • Audio unit terminal 1Q and Bluetooth® unit terminal Z (With Bluetooth® unit) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 11.
		No	Go to Step 9.
7	**INSPECT RELATED WIRING HARNESS (LINE-TO-LINE SHORT CIRCUIT)** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the following connectors: — Audio unit — CD player (With CD player) — Bluetooth® unit (With Bluetooth® unit) • Inspect between the following terminals (wiring harness-side) for continuity. — Audio unit terminal 1O and 1Q — CD player terminal G and H (With CD player) — Bluetooth® unit terminal Y and Z (With Bluetooth® unit) • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Audio unit terminal 1O and CD player terminal G (With CD player) • Audio unit terminal 1Q and CD player terminal H (With CD player) • Audio unit terminal 1O and Bluetooth® unit terminal Y (With Bluetooth® unit) • Audio unit terminal 1Q and Bluetooth® unit terminal Z (With Bluetooth® unit) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short between circuits. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has shorted between circuits. Go to Step 11.
		No	Go to Step 9.

Step	Inspection		Action
8	**INSPECT RELATED WIRING HARNESS (OPEN CIRCUIT)** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the following connectors: — Audio unit — CD player (With CD player) — Bluetooth® unit (With Bluetooth® unit) — Connector C-37 (L.H.D.) — Connector C-36 (R.H.D.) • Inspect between the following terminals (wiring harness-side) for continuity. — L.H.D.: • CD player terminal G and connector C-37 (With CD player) • CD player terminal H and connector C-37 (With CD player) • Bluetooth® unit terminal Y and connector C-37 (With Bluetooth® unit) • Bluetooth® unit terminal Z and connector C-37 (With Bluetooth® unit) • Audio unit terminal 1O and connector C-37 • Audio unit terminal 1Q and connector C-37 — R.H.D.: • CD player terminal G and connector C-36 (With CD player) • CD player terminal H and connector C-36 (With CD player) • Bluetooth® unit terminal Y and connector C-36 (With Bluetooth® unit) • Bluetooth® unit terminal Z and connector C-36 (With Bluetooth® unit) • Audio unit terminal 1O and connector C-36 • Audio unit terminal 1Q and connector C-36 • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • L.H.D.: — CD player terminal G and connector C-37 (With CD player) — CD player terminal H and connector C-37 (With CD player) — Bluetooth® unit terminal Y and connector C-37 (With Bluetooth® unit) — Bluetooth® unit terminal Z and connector C-37 (With Bluetooth® unit) — Audio unit terminal 1O and connector C-37 — Audio unit terminal 1Q and connector C-37 • R.H.D.: — CD player terminal G and connector C-36 (With CD player) — CD player terminal H and connector C-36 (With CD player) — Bluetooth® unit terminal Y and connector C-36 (With Bluetooth® unit) — Bluetooth® unit terminal Z and connector C-36 (With Bluetooth® unit) — Audio unit terminal 1O and connector C-36 — Audio unit terminal 1Q and connector C-36 **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 11.
		No	Go to Step 9.
9	**VERIFY IF MALFUNCTIONING LOCATION IS Bluetooth® UNIT DEPENDING ON REPEATABILITY** **Note** • For vehicles without the Bluetooth® unit, go to the next step. • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the Bluetooth® unit. (See CLEARING DTC [AUDIO].) • Perform the DTC inspection for the Bluetooth® unit. (See DTC INSPECTION [AUDIO].) • Is the same DTC displayed?	Yes	Replace the Bluetooth® unit, then go to the next step. (See Bluetooth® UNIT REMOVAL/INSTALLATION.)
		No	Go to the next step.

Step	Inspection		Action
10	**VERIFY IF MALFUNCTIONING LOCATION IS CD PLAYER DEPENDING ON REPEATABILITY** **Note** • For vehicles without the CD player, go to the next step. • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the audio unit DTCs. (See CLEARING DTC [AUDIO].) • Perform the audio unit DTC inspection. (See DTC INSPECTION [AUDIO].) • Is the same DTC displayed?	Yes	Replace the CD player, then go to the next step. (See CD PLAYER REMOVAL.) (See CD PLAYER INSTALLATION.)
		No	Go to the next step.
11	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the audio unit. (See CLEARING DTC [AUDIO].) • Perform the audio unit DTC inspection. (See DTC INSPECTION [AUDIO].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the audio unit. (See AUDIO UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
12	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [AUDIO].)
		No	DTC troubleshooting completed.